MAT137Y Calculus with Proofs

Proofs with Limits (1)

2021-2022 Fall Session

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2 Proving limits with arbitrary parameters

Given L as a constant, suppose

$$\lim_{x \to 0} f(x) = L$$

Prove

$$\lim_{x \to 0} f(2x) = L$$

This means we suppose

$$\forall \epsilon_2 > 0, \exists \delta_2 > 0, 0 < |x_2 - 0| < \delta_2 \Rightarrow |f(x_2) - L| < \epsilon_2$$

We need to prove

$$\forall \epsilon > 0, \exists \delta > 0, 0 < |x - 0| < \delta \Rightarrow |f(2x) - L| < \epsilon$$

2.1 Proving the above statement

Let $\epsilon > 0$

Show the existence of delta to make the statement hold. Unfortunately, you can't find the delta. However, we can pick a particular value of the quantified variable labeled under for every. Choose one that is advantageous to the proof.

Pick $\epsilon_2 = \epsilon$. Then

$$\exists \delta_2 > 0, \forall x_2 \text{ if } 0 < |x_2 - 0| < \delta_2 \Rightarrow \ |f(x_2) - L| < \epsilon_2$$

Now, ϵ_2 must always be picked prior to delta because the existence of delta depends on epsilon.

Pick $\delta =$ __ (leave that blank for now).

Assume $0 < |x - 0| < \delta$

Prove $|f(2x) - L| < \epsilon$

Consider (pick) $x_2 = 2x$. Why are we allowed to do this? Because we allowed the blue statement to be true, where $\forall x_2$ appears somewhere in the blue statement. Why did we choose x_2 to be that? Because it is advantageous to solving the problem. Now, the statement revisited:

$$\exists \delta_2 > 0, \forall x_2 \text{ if } 0 < |x_2 - 0| < \delta_2 \Rightarrow \ |f(x_2) - L| < \epsilon_2$$

Because we picked $x_2 = 2x$, the red statement above becomes the statement we need to prove, as $|f(x_2) - L| = |f(2x) - L|$. How do we prove it? We need to prove the green statement.

When you want to prove an implication, you assume the statement after the "if" and prove the statement after the then.

However, **when you know an implication is true,** and you want to use it, you actually prove the statement after the "if" and then conclude the statement after the then. You do not prove

the statement after the "then" because we already know the implication is true. This means *if the green statement is true, the red statement is true. Proving that the green statement is true verifies the red statement.*

Ask: $0 < |x_2 - 0| < \delta_2$. Can we get this to hold (what to solve, given $|x - 0| < \delta$)?

Since (scroll back up to our assumption), we can end up with:

$$\begin{aligned} &|x - 0| = |x| < \delta \\ &0 < |2x| < 2\delta \\ &0 < |x_2 - 0| < 2\delta \end{aligned}$$

However, we want to end up with $|x_2 - 0| < \delta_2$, not 2δ. This is where we can take advantage of our "pick $\delta =$ ___." Remember that we are free to pick δ to be whatever we want, and if we pick it to be $\delta = \frac{\delta_2}{2}$, then $\delta_2 = 2\delta$. Now we can substitute:

$$0 < |x_2 - 0| < \delta_2$$

Once we know that this (above) is true, we can also conclude (based on our assumption) that $|f(x_2) - L| < \epsilon_2$ is true.

Because of our choices that $\epsilon_2 = \epsilon$ and $x_2 = 2x$:

$$|f(x_2) - L| < \epsilon_2 \Leftrightarrow |f(2x) - L| < \epsilon$$

■

2.1.1 Shortened

Suppose

$$\forall \epsilon_2 > 0, \exists \delta_2 > 0, 0 < |x_2 - 0| < \delta_2 \Rightarrow |f(x_2) - L| < \epsilon_2$$

We need to prove

$$\forall \epsilon > 0, \exists \delta > 0, 0 < |x - 0| < \delta \Rightarrow |f(2x) - L| < \epsilon$$

Proof. Let $\epsilon > 0$. Pick $\epsilon_2 = \epsilon$, pick $\delta = \frac{\delta_2}{2}$.

Assume $|x - 0| < \delta$. Pick $x_2 = 2x$. This means

$$\begin{aligned} 2|x - 0| &< 2\delta \\ |2x - 0| &< 2\delta \\ |x_2 - 0| &< 2\delta \\ |x_2 - 0| &< \delta_2 \end{aligned}$$

This implies that $|f(x_2) - L| < \epsilon_2$ and also that $|f(2x) - L| < \epsilon$.

■

2.2 Proving the sum of two limits

$|f(x) + g(x) - (L + M)| < \epsilon$

Triangle inequality: While you can split absolute values over multiplications, they can't exactly be split when doing addition. $|x+y| \leq |x| + |y|$

Proving the triangle inequality

Assume your inequality to be true (but don't write it) (start with the inequality you want to prove and do some algebra, simplify it, and end up with something that is necessarily true).

What is $|a|^2$? It's a^2. If a was negative and you square it, the negative would go away anyways. Use the square operator to remove the absolute value.

$$\begin{gathered}
|x+y| \leq |x| + |y| \\
|x+y|^2 \leq (|x| + |y|)^2 \\
(x+y)^2 \leq (|x| + |y|)^2 \\
x^2 + 2xy + y^2 \leq |x|^2 + 2|x||y| + |y|^2 \\
x^2 + 2xy + y^2 \leq x^2 + 2|x||y| + y^2 \\
2xy \leq 2|x||y| \\
xy \leq |xy|
\end{gathered}$$

Is the absolute value of a number always greater than or equal to that number? Yes. $xy \leq |xy|$ is true beyond reasonable doubt.

But why does using algebra help prove the statement? Because after each algebraic step you are making a double-sided implication, where each step implies and are equivalent to each other.

2.2.1 Proof

We assumed the following

$$\begin{gathered}
\forall \epsilon_2 > 0, \exists \delta_2 > 0, 0 < |x_2 - 0| < \delta \Rightarrow |f(x_2) - L| < \epsilon_2 \\
\forall \epsilon_3 > 0, \exists \delta_3 > 0, 0 < |x_3 - 0| < \delta \Rightarrow |g(x_3) - M| < \epsilon_3
\end{gathered}$$

Prove $\lim_{x \to 0} f(x) + g(x) = L + M$

Prove $\forall \epsilon > 0$ if $0 < |x - a| < \delta \Rightarrow |f(x) + g(x) - (L + M)| < \epsilon$

Let $\epsilon > 0$ be arbitrary.

Pick $\epsilon_2 = \frac{\epsilon}{2}$, (we wouldn't know what ϵ_2 is, but it is whatever we want any time in the future) and assume the bottom statement to be true:

$$\exists \delta_2 > 0 \text{ if } 0 < |x_2 - a| < \delta_2 \Rightarrow |f(x_2) - L| < \epsilon_2$$

Pick $\epsilon_3 = \frac{\epsilon}{2}$, and assume the statement below to be true:

$$\exists \delta_3 > 0 \text{ if } 0 < |x_3 - a| < \delta_3 \Rightarrow |g(x_3) - M| < \epsilon_3$$

Pick $\delta = \min\{\delta_2, d_3\} \Rightarrow \delta \leq \delta_2, \delta \leq d_3$

Prove what we need to solve: assume the "if" and prove the "then."

Assume $0 < |x - a| < \delta$

Prove $|f(x) + g(x) - (L + M)| < \epsilon$

Consider $x_2 = x, x_3 = x$.

Since $0 < |x - a| < \delta$ (we're using the picked blue statement to justify our substitution):

$$\begin{aligned} 0 \leq |x - a| = |x_2 - a| < \delta \leq \delta_2 \\ 0 < |x - a| = |x_3 - a| < \delta \leq \delta_3 \end{aligned}$$

Which helps up conclude:

$$\begin{aligned} &\Rightarrow |f(x_2) - L| < \epsilon_2 \\ &\Rightarrow |g(x_3) - M| < \epsilon_3 \end{aligned}$$

$$\begin{aligned} &|f(x) + g(x) - (L + M)| \\ &= |(f(x) - L) + (g(x) - M)| \\ &\leq |f(x) - L| + |g(x) - M| \\ &< \epsilon_2 + \epsilon_3 = \epsilon \end{aligned}$$

In the process, we picked $\epsilon_2 = \frac{\epsilon}{2}$ and $\epsilon_3 = \frac{\epsilon}{2}$, as we wanted to end our proof with epsilon.

We thus conclude $|f(x) + g(x) - (L + M)| < \epsilon$.

■

3 Using existing statements to prove other statements

($\wedge$ means "and"; $\vee$ means "or".)

A statement is **necessarily true** means we can prove that statement given using the assumption.

A statement is **not necessarily true** (considered false, usually the result of not all $\forall$ checks passing despite some) when there is a counterexample.

3.1.1 Format

Knowing

$$A \Rightarrow B$$

Prove that

$$C \Rightarrow D$$

1. Assume C is true.
2. Try to prove A is true, then conclude B is true for this context.
3. Use B to prove D or find a counterexample.

The end goal is to prove D is true and C is false. Remember that $C \Rightarrow D$ is logically equivalent to $\neg C \vee D$.

For example, if we know this is true (statement 1 a.k.a. first equation, from now on):

$$|x-2| < 1 \Rightarrow |f(x)-5| < 2$$

3.1.2 Example 1

We can prove the following:

$$|x-2| < 1 \Rightarrow |f(x)-5| < 3$$

Here's the rationale: If we assume $|x-2| < 1$, we know that $|f(x)-5| < 2$. Because $2 < 3$, $|f(x)-5| < 2 \Rightarrow |f(x)-5| < 3$ and therefore what we wanted to prove at the start is proven.

3.1.3 Example 2

Another statement to prove using the fact that statement 1 is true:

$$|x-2| < 0.5 \Rightarrow |f(x)-5| < 2$$

By assuming $|x-2| < 0.5$, we can conclude $|x-2| < 1$, which immediately implies $|x-2| < 0.5$ and $|f(x)-5| < 2$ (from the first statement, which turns out to be what we needed to prove true for the statement we needed to prove.

3.1.4 Example 3

$$|x-2| < 1 \Rightarrow |f(x)-5| < 1$$

Assume $|x-2| < 1$. This means $|f(x)-5| < 2$, but can we conclude $|f(x)-5| < 1$? No. Think about it. For example, if $3 < f(x) < 7$, then $|f(x)-5| < 1 \wedge |f(x)-5| < 2$, but if we let $f(x) = 7.5$, $|7.5-5| = 1.5 < 2$, but $1{,}5 \not< 1$. This is the reasoning why the statement we needed to prove is not necessarily true. In other words, the steps to prove/disprove a statement like this is:

1. Assume "if" of what to solve
2. If the "if" of what to solve implies the "if" of the first equation, then we can guarantee the "then" of the first equation is true.
3. The "then" of what to solve must be implied by the "then" of the first equation. You can also find a counterexample to disprove a statement.

A negation of an implication is where the "if" is true and the "then" is false, always.

Consider $x = 2$, and $f(x) = 6.5$. ($f(x)$ can be anything, as we did not put any constraints on it.)

$$|x-2| = 0 < 1 \Rightarrow |f(x)-5| = |6.5-5| \not< 1$$

(For what we know that is true must be followed when constructing the above work.)

3.1.5 Example 4

Knowing that:

$$|x-2| < 1 \Rightarrow |f(x)-5| < 2$$

Prove:

$$|x-2| < 0.5 \Rightarrow |f(x)-5| < 1$$

Proof.

Assume $|x-2| < 0.5$; this implies $|x-2| < 1$, and thus implies $|f(x)-5| < 2$. If you scroll up, you can find why $|f(x)-5| < 1$ is false (not always true). Do this by giving an example where $|x-2| < 0.5$ is true and $|f(x)-5| < 1$ is false. Scroll up to question 3 for when that is used.

3.1.6 Example 5

Knowing that

$$|x-2| < 1 \Rightarrow |f(x)-5| < 2$$

Prove

$$\exists \delta > 0, |x-2| < \delta \Rightarrow |f(x)-5| < 1$$

To show WTS is false, show

$$\neg(\exists \delta > 0, |x-2| < \delta \Rightarrow |f(x)-5| < 1)$$
$$\forall \delta > 0, |x-2| < \delta \wedge |f(x)-5| \not< 1$$

(A very small delta will imply that x can only be 2; any greater than it will exceed δ)

Let $\delta > 0$ be arbitrary. Consider $x = 2$. This means $|x-2| < \delta$. (Why did we pick $x = 2$? Because it is the only x that makes $|x-2| < \delta$ where delta can be as close to 0 as it wants but never negative)

Consider $f(x) = 6.5$

It satisfies $|x-2| = 0 < 1 \Rightarrow |f(x)-5| < 2$

With that checked off, the value we assigned to $f(x)$ is legal. Then $|6.5-5| = 1.5 \not< 1$.

This is satisfied, ultimately proving that

$$\forall \delta > 0, |x-2| < \delta \wedge |f(x)-5| \not< 1$$

Is true.

4 Proving that a limit doesn't exist

We still have the assumption

$$|x-2| < 1 \Rightarrow |f(x)-5| < 2$$

We know something about f, but we don't know what the function truly is. It is just the function that satisfies the assumption.

Now we suppose that

$$\lim_{x \to 2} f(x) = L$$

This means, we know the following is true:

$$\forall \epsilon > 0, \exists \delta > 0, 0 < |x - 2| < \delta \Rightarrow |f(x) - L| < \epsilon$$

4.1 Deciding whether to prove the limit exists or not

Can $L = 2$? Give an example of a function with that limit being two. If you think the limit cannot be two, then prove it using the assumption.

If you say so, it means $L - \epsilon < f(x) < L + \epsilon \Rightarrow f(x) = 2$

Now what do we know about our first assumption? Then $|f(x) - 5| < 2 \Leftrightarrow 3 < f(x) < 7$

The two above statements contradict. L cannot be 2. A.K.A. because $f(x) \in (3, 7)$ and we wanted $f(x)$ to output 2, but we know that can't be the case because $f(x)$ is restricted.

To prove an implication is false: A is true, and B is false.

4.2 The "Proving the limit doesn't exist"

Prove $\lim_{x \to 2} f(x) \neq 2$. To prove a limit is not something, negate the limit definition. So, the limit definition, negated is:

$$\exists \epsilon > 0, \forall \delta > 0, \exists x, 0 < |x - 2| < \delta \wedge |f(x) - L| \geq \epsilon$$

Pick $\epsilon = 0.5$ (Initially blank and to be written down later. Could be small as we want.)

Let $\delta > 0$ be randomly chosen.

Pick $x = 2 + \min\{\frac{\delta}{2}, \frac{1}{2}\}$ (Initially blank. You technically can pick $2 + \frac{\delta}{69696969}$ but please don't.)

Prove $0 < |x - 2| < \delta \wedge |f(x) - L| \geq \epsilon$

Firstly, prove that the left side is true. Pick a particular x that makes the blue statement is true. Remember: our assumption still holds:

$$\begin{aligned} &|x - 2| < 1 \Rightarrow |f(x) - 5| < 2 \\ &\Rightarrow 1 < x < 3 \\ &\Rightarrow 3 < f(x) < 7 \end{aligned}$$

Okay, back to $0 < |x - 2| < \delta$. Notice that $x \neq 2$. 2 is the most obvious answer, but there is a slight technical detail that needs to be fixed. The statement $0 < |x - 2| < \delta \Rightarrow 2 - \delta < x < 2 + \delta$. We know δ can be really small, but not at zero. Therefore, make x be slightly larger than 2. This is why we picked $x = 2 + \frac{\delta}{2}$, which was picked just now.

Okay, how would you know that $1 < 2 + \frac{\delta}{2} < 3$? δ could be anything. The way to solve this is to use the minimum function. This means $x = 2 + \min\{\frac{\delta}{2}, \frac{1}{2}\}$.

Now since we chose x to be that:

$$x \leq 2 + \frac{\delta}{2} < 2 + \delta$$
$$x \leq 2 + \frac{1}{2} < 3 \Leftrightarrow x \leq 2.5 < 3$$

Compress this, and we have $x \leq 2 + \frac{\delta}{2}$ and $x \leq \frac{5}{2}$.

Since $\frac{\delta}{2} > 0$ and $\frac{1}{2} > 0$, $x > 2$ if we pick either of them. (x is designed to be ever so slightly greater than two.)

Combining the statements, we wrote above: $2 < x < 2 + \delta$

$$\Rightarrow 2 - \delta < x < 2 + \delta$$
$$\Leftrightarrow 0 < |x - 2| < \delta$$

What we know: This limit exists. $\lim_{x \to 2} f(x) = L$. Can the limit be equal to two? Prove that it can't, which is what we want to prove.

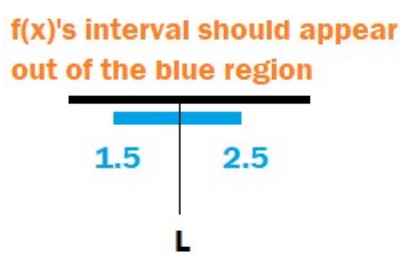
Back to what we needed to prove: $0 < |x - 2| < \delta \wedge |f(x) - L| \geq \epsilon$. We proved $0 < |x - 2| < \delta$, and now we need to prove $|f(x) - L| \geq \epsilon$.

These are the assumptions we have in our disposal:

$$1 < x < 3 \Rightarrow 3 < f(x) < 7$$

We can confirm that we did declare x to be slightly greater than two and we know that it is less than 3 (scroll up), so we can confirm $2 < x < 3 \Rightarrow 1 < x < 3$

Now we finally know $3 < f(x) < 7$ is true. Knowing this is the fact, we need to prove $|f(x) - L| \geq \epsilon$, which is **equivalent** to $f(x) \geq L + \epsilon, f(x) \leq L - \epsilon$ **or** $f(x) \notin (L - \epsilon, L + \epsilon)$ – if this statement meets for all intervals of $f(x)$, **then we proved what we needed to solve:** $|f(x) - L| \geq \epsilon$ (Recall $L = 2$)



We didn't choose what epsilon is yet, so can we choose an epsilon in a way that the interval of $3 < f(x) < 7$ lies outside of the diagram above? Yes. Just let $\epsilon = 0.5$. We know L is 2, and that guarantees the interval of $f(x)$ is outside the interval of $(L - \epsilon, L + \epsilon)$.

Therefore $f(x)$ does not lie in the interval $(1.5, 2.5)$

A diagram that visualizes it: $f(x)$ cannot be within the blue zone.

$\square$

4.2.1 Formal Proof, shortened

Suppose the following are true:

$$|x-2| < 1 \Rightarrow |f(x)-5| < 2$$
$$\lim_{x\to 2} f(x) = L$$

Prove $L \neq 2$.

From the limit presented, we get this, which we can assume holds.

$$\forall \epsilon > 0, \exists \delta > 0, 0 < |x-2| < \delta \Rightarrow |f(x) - L| < \epsilon$$

This is an implication that proves a limit exists. We want to prove that the limit doesn't exist, so we reverse the epsilon-delta definition of the limit $\lim_{x\to 2} f(x) = L$:

$$\exists \epsilon > 0, \forall \delta > 0, 0 < |x-2| < \delta \wedge |f(x) - L| \geq \epsilon$$

Our task is to prove this with $L = 2$.

Pick $\epsilon = 0.5$

Let $\delta > 0$ be picked arbitrarily / randomly chosen.

Pick $x = \min\{2 + \frac{\delta}{2}, 2 + \frac{1}{2}\}$

Prove $0 < |x-2| < \delta \wedge |f(x) - L| \geq \epsilon$

We want x to be a value slightly greater than 2. So, we pick $x = \min\{2 + \frac{\delta}{2}, 2 + \frac{1}{2}\}$ (The right side of the minimum function prevents x from being 3 or greater as it breaks one of our assumptions.

This means $x \leq 2 + \frac{\delta}{2}$ and $x \leq 2 + \frac{1}{2}$. The minimum function also tells us $x > 2$ as $\frac{\delta}{2}, \frac{1}{2} > 0$. We can thus write: (also $2 + \frac{\delta}{2} < 2 + \delta$)

$$2 < x < 2 + \delta$$
$$\Leftrightarrow 0 < |x-2| < \delta$$

We proved the left side of $0 < |x-2| < \delta \wedge |f(x) - L| \geq \epsilon$. Now to prove $|f(x) - L| \geq \epsilon$

Remember our assumptions: $|x-2| < 1 \Rightarrow |f(x) - 5| < 2$

$$(|x-2| < 1 \Leftrightarrow 1 < x < 3) \Rightarrow (|f(x) - 5| < 2 \Leftrightarrow 3 < f(x) < 7)$$

It means $f(x)$ must be between 3 and 7, non-inclusively. Now, we let $L = 2$, remember? This means:

$$|f(x) - 2| \geq \epsilon$$
$$f(x) \geq 2 + \epsilon \vee f(x) \leq 2 - \epsilon$$

We said we could let ϵ be anything, so let's make it 0.5; this is what happens:

$$f(x) \geq 2.5 \vee f(x) \leq 1.5$$

$$2.5 \leq 3 < f(x) < 7$$

(This says $f(x)$ must be greater that 2.5 or something else. But $f(x)$ has to be between 3 and 7. This means $f(x)$ must be greater than 2.5; statement proved – we proved the other side of the "and".)

4.2.2 Shortened, again

$$|x - 2| < 1 \Rightarrow |f(x) - 5| < 2$$
$$\lim_{x \to 2} f(x) = L$$

$$\exists \epsilon > 0, \forall \delta > 0, 0 < |x - 2| < \delta \wedge |f(x) - L| \geq \epsilon$$

Pick $\epsilon = 0.5$, let $\delta > 0$ be picked arbitrarily / randomly chosen, and pick $x = \min\{2 + \frac{\delta}{2}, 2 + \frac{1}{2}\}$

Prove $0 < |x - 2| < \delta$. This statement is the same as $2 < x < 2 + \delta$. Because $x = \min\{2 + \frac{\delta}{2}, 2 + \frac{1}{2}\}$, then $2 < x$ and $x < 2 + \delta$. We can thus verify the statement.

Prove $|f(x) - L| \geq \epsilon$. Choose $\epsilon = \frac{1}{2}$. Then we need to prove that $f(x)$ is not in $(L - \epsilon, L + \epsilon)$. This means $f(x)$ is not in $(2 - 0.5, 2 + 2.5)$. We can prove that is the case because $f(x)$ is assumed to be in the interval $(3, 7)$. Therefore, proven.

4.3 Preliminary conclusions

We know $0 < |x - 2| < \delta$ is true, as L does not affect it.

$L = 2.5$? Choose $\epsilon = \frac{1}{4}$. Then we need to prove that $f(x)$ is not in $(L - \epsilon, L + \epsilon)$. This means $f(x)$ is not in $(2.25, 2 + 2.75)$. We can prove that is the case because $f(x)$ is assumed to be in the interval $(3, 7)$. Therefore, proven.

$L = 8$? Choose $\epsilon = \frac{1}{2}$. Then we need to prove that $f(x)$ is not in $(L - \epsilon, L + \epsilon)$. This means $f(x)$ is not in $(8 - 0.5, 8 + 0.5)$. We can prove that is the case because $f(x)$ is assumed to be in the interval $(3, 7)$. Therefore, proven.

We can conclude preliminary (by realizing) that $L \notin (3, 7) \Rightarrow$ limit does not exist. However, we have only checked the values of values out of $[3, 7]$.

$L = 5$? This means $3 < f(x) < 7$ has to hold. Therefore, we think that it is possible, but it is not a proper answer, as we need to give an example $f(x)$ with $L = 5$, so we can have $f(x) = 5$: it satisfies $3 < f(x) < 7$.

Suppose $|x-2|<1$, prove $|f(x)-5|<2$. $|f(x)-5| \Rightarrow |f(x)-5| = |5-5| < 2$; therefore, we can be confident that $L=5$ is a possibility, but not completely.

Consider $L=4$. Suppose $|x-2|<1$, prove $|f(x)-4|<2$. $|f(x)-5| \Rightarrow |4-5| = |4-5| < 2$; therefore, we can be confident that $L=4$ is a possibility, but not completely.

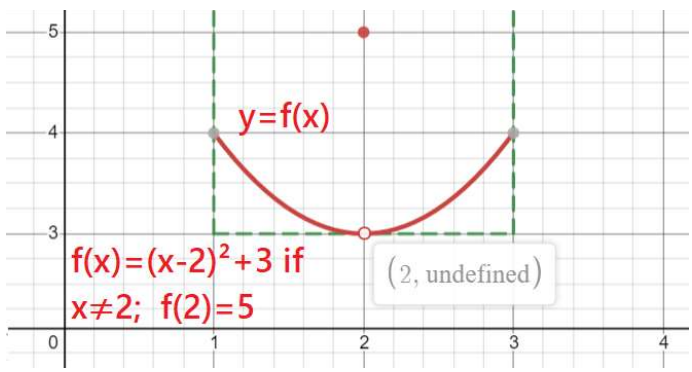
4.4 It is not always what it seems

$$|x-2|<1 \Rightarrow |f(x)-5|<2$$
$$\lim_{x\to 2} f(x) = L$$

Remember the definition of a limit at $x=2$:

$$\forall \epsilon > 0, \exists \delta > 0, 0 < |x-2| < \delta \Rightarrow |f(x)-L| < \epsilon$$

Can $L=3$? We cannot make the conclusion that $L=3$ is impossible, even though making a preliminary conclusion makes it look like $L=3$ is impossible. Suppose $|x-2|<1$, prove $|f(x)-5|<2$. $|f(x)-5| \Rightarrow |3-5| = |3-5| < 2 \Rightarrow 2 < 2$... however, that does not rule out the possibility of $L=3$. If we let $f(x)$ be a function represented by the black line with a hole at $x=2$, the graph $y=f(x)$ approaches 3 from above but doesn't hit it.



Proof.

Consider $f(x) = \begin{cases} (x-2)^2+3, x \neq 2 \\ 5, x = 2 \end{cases}$. The limit of $f(x)$ at 2 would be 3, but $f(2)=5$. That point (in red) is irrelevant when we consider limits. We have a specific condition that $x \neq 2$. So:

Suppose $|x-2|<1$ (an arbitrary value between 1 and 3). Prove $|f(x)-5|<2$.

Case 1: $x=2$. Then $f(x)=5$, and $|f(x)-5| = |5-5| < 2$, sastifying that $|f(x)-5|<2$

Case 2: $x \neq 2$. Since $|x-2|<1$

$$0 < |x-2|^2 < 1^2$$
$$0 < (x-2)^2 < 1$$
$$3 < (x-2)^2+3 < 4$$
$$3 < f(x) < 4$$

$$-2 < f(x) - 5 < -1$$
$$1 < |f(x) - 5| < 2$$

This proves satisfied that this function works. This means L can be 3, and we can also come up that $3 \leq f(x) \leq 7$ (though not proven yet).

5 Limits and Continuity

Sided-limits are only meant for 2-dimensional visualizations and can be used for computations.

5.1 Two-sided limits

The **limit** at a point a of the function $f(x)$ is equal to the number L.

1. When you let x approach a from the left side **and** the right side of a, the graph of $f(x)$ gets close to number L.
2. The actual value of $f(a)$ does **NOT** matter at all.

$$\lim_{x \to a} f(x) = L$$

5.2 One-sided limits

A limit from the left side is $\lim_{x \to a^-} f(x) = L_1$. A limit from the right side is $\lim_{x \to a^+} f(x) = L_2$. If the two values match, the limit of $f(x)$ at a from both sides exist.

The formal definition of a left side limit is if $0 < a - x < \delta$, and the right limit is $a < x < a + \delta$. These two cannot be generalized for higher dimensions.

5.3 Continuity

Intuitively, $f(x)$ is continuous in an interval if the graph if the graph is in one piece. $f(x)$ is continuous at the point a if the limit a is equal to $f(x)$:

$$\lim_{x \to a} f(x) = f(a)$$

We can only define that a function is continuous on a point, but using that we can:

$f(x)$ is **continuous** in an interval means $f(x)$ is continuous at every point in the interval. Most functions are continuous. Transforming a continuous function using any means (such as $+$, $\times$, and composition) does not change whether the transformed interval is continuous (except division by zero.) Every standard function we know are continuous in their domains.

E.g., $\lim_{x \to 1}(x^2 + 2^x) = f(1) = 1^2 + 2^1 = 3$

Find conditions on A and B so that

$$f(x) = \begin{cases} Ax - B & x \leq 1 \\ 3x & 1 < x < 2 \\ Bx^2 - A & 2 \leq x \end{cases}$$

Is continuous at $x = 1$ and discontinuous at $x = 2$.

Continuous: This means $\lim_{x \to 1} f(x) = f(1)$ and $\lim_{x \to 2} f(x) \neq f(2)$. Fortunately, all of the pieces of the piecewise functions are continuous. To evaluate this question, we need to use left and right sided limits.

$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} Ax - B = A - B$$

$$\lim_{x \to 1^+} f(x) = \lim_{x \to 1^+} 3x = 3x = 3$$

$$A - B = 3 = \lim_{x \to 1} f(x) = f(1)$$

As long $A - B = 3$, then $f(x)$ is continuous at 1.

Discontinuous: This means $\lim_{x \to 2} f(x) \neq 2$

$$\lim_{x \to 2^-} f(x) = \lim_{x \to 2^-} 3x = 6$$

$$\lim_{x \to 2^+} f(x) = \lim_{x \to 2^+} (Bx^2 - A) = 4B - A$$

Now, we need to make sure $\lim_{x \to 2} f(x) \neq f(2)$. If this two-sided limit were to exist, the left limit and the right limit would match, meaning $6 = 4B - A = \lim_{x \to 2} f(x) = f(2)$. When would the limit not exist? $6 \neq 4B - A$. Solve it.

$$\begin{cases} 6 \neq 4B - A \\ 3 = A - B \end{cases}$$

5.4 Limits approaching infinity

We allow a to be infinite as well. This is understood intuitively as:

As x becomes larger and largher, $f(x)$ gets closer to a number L.

$$\lim_{x \to \infty} f(x) = L$$

We can also have a being minus infinity:

$\lim_{x \to -\infty} f(x) = L$

5.5 Limit Properties

First, we **need** to know the following limits to exist:

$$\lim_{x \to a} f(x) = L \,, \lim_{x \to a} g(x) = M$$

If we are not certain these limits exist, we cannot use the properties below.

Then we can attain

$$\lim_{x\to a}[f(x)+g(x)] = L+M$$

$$\lim_{x\to a}[f(x)-g(x)] = L-M$$

$$\lim_{x\to a} c\cdot f(x) = c\cdot L$$

$$\lim_{x\to a}[f(x)\cdot g(x)] = L\cdot M$$

$$\lim_{x\to a}\frac{f(x)}{g(x)} = \frac{L}{M}, M\neq 0$$

Third property onward, proving these become extremely hard.

If $f(x)$ is continuous, we can move the limit **inside** f:

$$\lim_{x\to a} f\big(g(x)\big) = f\left[\lim_{x\to a} g(x)\right]$$

5.5.1 Limit properties proofs

If we know two limits is true

Firstly, we have this statement brought from the previous subsection:

$$\lim(f)\ \text{ and}\lim(g)\ \text{ exists} \Rightarrow \lim(f+g) = \lim(f)+\lim(g) \tag{1}$$

Prove the following statement: Suppose $\lim_{x\to a}[f(x)+g(x)]$ exists and $\lim_{x\to a} f(x)$ exists, then $\lim_{x\to a} g(x)$ also exists. Unfortunately, it is not correct to use (1) at first. You need to prove (1) first, so we know that it holds, and for (1) to be true we need to know that the limit at f and g exists. However, we do not know for sure that $\lim(g)$ exists, meaning we **cannot assume that (1) is true.**

What would happen if we considered $\lim_{x\to a}(f+g) - \lim_{x\to a} f$. Are we allowed to consider this? This requires us to use this property:

$$\lim(f)\ \text{ and}\lim(g)\ \text{ exists} \Rightarrow \lim(f+g)\lim(f-g) = \lim(f)-\lim(g) \tag{2}$$

This doesn't hold. We do not know that $\lim_{x\to a} g(x)$ exists.

What if we consider the function $f+g$ and the function f? We know that $\lim(f+g)$ exists and so is $\lim(f)$, so we can deduce that $\lim\big((f+g)-f\big)$.

The key point is to use limit laws, you need to know that the two limits you would use the laws exist in the first place. We considered $\lim(f+g)$ to be one function, and $\lim(f)$ to be the other, which is why it is perfectly acceptable to write $\lim(f+g-f)$. Is this confusing? Think as if I made the substitution $\lim(f+g) = \lim(h)$.

We have $\lim(f+g-f) = \lim(g)$, **which we know exists** as it was derived from a limit that exists. It is precise to say:

Since $\lim(f+g)\,,\lim(f)$ *exists, then we can conclude that* $\lim(f+g)-\lim(f)$ *exists, and so on.*

Part 2

T/F? Suppose $\lim\limits_{x\to a}[f(x)+g(x)]$ exists but $\lim\limits_{x\to a} f(x)$ does not exist, then $\lim\limits_{x\to a} g(x)$ also does not exist.

Proof. Suppose $\lim\limits_{x\to a} g(x)$ exists. We know $\lim\limits_{x\to a}[f(x)+g(x)]$ exists.

Consider $\lim(f+g)-\lim(g) \Rightarrow \lim(f+g-g) \Rightarrow \lim(f)$ (we can do this because the limit of $f+g$ and g are assumed to exist), meaning we proved that $\lim(f)$ exists. However, we have reached a statement which is contradictory to something we assumed initially. This means it is not possible for $\lim(g)$ to exist.

When you reach a contradiction, your "supposed" assumption will automatically falsify itself (or turn true if supposed false). For regular proof questions, we cannot assume what we want to prove. However, when working with true/false questions, take a guess and go both paths.

Part 3

Suppose $\lim\limits_{x\to a} f(x)$ exists but $\lim\limits_{x\to a} g(x)$ does not exist, then $\lim\limits_{x\to a}[f(x)+g(x)]$ does not exist. This statement is true. $\lim\limits_{x\to a}[f(x)+g(x)]$ cannot exist if either limit of f or g do not exist – the limit property $\lim\limits_{x\to a}[f(x)+g(x)]$ only works if the two functions as individual limits work in the first place. Here, we know the opposite.

5.6 Continuity law for composition

Prove the *continuity law for composition:*

If $f(x)$ is continuous at $\lim\limits_{x\to a} g(x)$, then we can move the limit inside:

$$\lim_{x\to a} f\big(g(x)\big) = f\left[\lim_{x\to a} g(x)\right] \tag{1}$$

Deducing this implies that composition of continuous functions is continuous.

We know that if you put a continuous function inside another continuous function, you get a composite function. However, we think of continuous as a graph being in one piece. However, you have no idea.

What does it mean for $f(x)$ to be continuous? It means $\lim\limits_{x\to a} f(x) = f(a)$. How would we write this statement?

$$\forall \epsilon > 0, \exists \delta > 0, 0 < |x-a| < \delta \Rightarrow |f(x) - f(a)| < \epsilon$$

Because $L = f(a)$, we modified our epsilon-delta definition accordingly. Our assumption is that $f(x)$ is continuous at $\lim_{x\to a} g(x)$, but that should be treated just like a number. This means, which we know is true:

$$\lim_{x\to \lim_{x\to a} g(x)} f(x) = f\left[\lim_{x\to a} g(x)\right] \tag{2}$$

This is the true definition of if $f(x)$ is continuous at $\lim_{x\to a} g(x)$ (we know $\lim_{x\to a} g(x)$ exists, because we must assume everything in the assumption to be true). This is what the question is asking.

Now, what does (1) mean? **WTS:**

$$\forall \epsilon > 0, \exists \delta > 0, 0 < |x-a| < \delta \Rightarrow \left|f\big(g(x)\big) - f\left[\lim_{x\to a} g(x)\right]\right| < \epsilon \tag{3, To show}$$

This is a modified $\epsilon - \delta$ definition presented in the box if you scroll up. This statement is what (1) is trying to say. All we need to do is prove that (3) is true. This is where we start using our assumptions. Meaning we know

$$\forall \epsilon_2 > 0, \exists \delta_2 > 0, 0 < \left|x_2 - \lim_{x\to a} g(x)\right| < \delta_2 \Rightarrow \left|f(x_2) - f\left(\lim_{x\to a} g(x)\right)\right| < \epsilon_2 \tag{4, True}$$

We used assumption (2) to help us construct the limit definition (4) that we can assume true when proving (3). Our (4) is something that we use.

Proof. Let $\epsilon > 0$. Pick $\epsilon_2 =$ __, which makes the statement below true.

$$\exists \delta_2 > 0, 0 < \left|x_2 - \lim_{x\to a} g(x)\right| < \delta_2 \Rightarrow \left|f(x_2) - f\left(\lim_{x\to a} g(x)\right)\right| < \epsilon_2 \tag{5, True}$$

Pick $\delta =$ __. It may depend on δ_2.

From (4), suppose $0 < |x-a| < \delta$

Prove $\left|f\big(g(x)\big) - f\left[\lim_{x\to a} g(x)\right]\right| < \epsilon$.

We haven't started the proof yet. We have set up the proof, though.

So, we pick $\epsilon_2 = \epsilon$ and $x_2 = g(x)$. Then $\left|f(x_2) - f\left(\lim_{x\to a} g(x)\right)\right| < \epsilon_2$ from (4) becomes $\left|f\big(g(x)\big) - f\left(\lim_{x\to a} g(x)\right)\right| < \epsilon_2$ from (5.0). This means we know this is true:

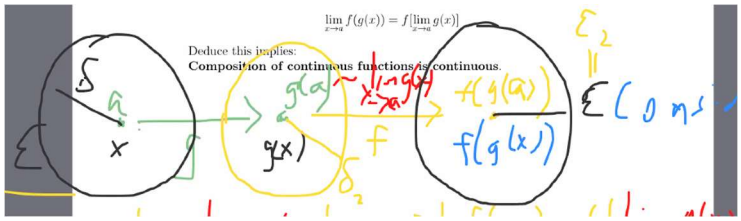
$$\forall \epsilon_2 > 0, \exists \delta_2 > 0, 0 < \left|g(x) - \lim_{x\to a} g(x)\right| < \delta_2 \Rightarrow \left|f\big(g(x)\big) - f\left(\lim_{x\to a} g(x)\right)\right| < \epsilon \tag{6, True}$$

(We are so close! Verify $0 < \left|g(x) - \lim_{x\to a} g(x)\right| < \delta_2$)

Connection: If $a \to g(a) \approx \lim_{x\to a} g(x) \to f\big(g(a)\big)$

If I pick an x that is very close to a, then $x \to g(x) \to f\big(g(x)\big)$. We are asking if x was near a, can we prove $f\big(g(x)\big)$ would be very close to $f\big(g(a)\big)$? We then pick an $\epsilon > 0$ which gives us an error range around $f\big(g(a)\big)$, and we would like to ask: can we find a δ range on the left. Then $f\big(g(x)\big)$ will be in the epsilon range within $f\big(g(a)\big)$? Then notice that if you start with x, if you

want to get to the right side, it needs to get to $g(x)$, then $g(x)$ would be inputted to f to obtain $f\big(g(x)\big)$. Now, $\exists\delta_2$ around $g(a)$, then $f\big(g(x)\big)$ would be very close to $f\big(g(a)\big)$. If we take a look at how δ_2 is set up, the delta we are looking for is all the way on the left. They are not even in the same place.



There is an implicit assumption that $\lim_{x\to a} g(x) = M$ exists. Then, we know that

$$\forall\epsilon_3 > 0, \exists\delta_3 > 0, 0 < |x_3 - a| < \delta_3 \Rightarrow \left|g(x_3) - \lim_{x\to a} g(x)\right| < \epsilon_3 \tag{7, True}$$

Pick $\epsilon_3 = \delta_2$. Then

$$\exists\delta_3 > 0, 0 < |x_3 - a| < \delta_3 \Rightarrow \left|g(x_3) - \lim_{x\to a} g(x)\right| < \epsilon_3$$

Pick $\delta = \delta_3$.

Now how can I make sure $g(x)$ lands in the δ_2 range?

Using the red statement: Give me any ϵ_3 range in the middle centered around the red point, I will make sure $g(x)$ lands in the ϵ_3 error range in the middle by giving a δ_3 control on the left side. This is why δ_2 is the control of the f function, which is the target of the g function. This justifies why we picked those variables. These guarantees $g(x)$ will be sent to $f\big(g(x)\big)$. It is not that these choices are intuitive.

1. Set a target on the right
2. Want a control on the left
3. Two-leg journey broken up in two pieces
4. Have $g(x)$ land in $\delta_2 = \epsilon_3$ which grants a δ_3 control range on the left
5. That will guarantee $f\big(g(x)\big)$ will land on ϵ.

Considering $x_2 = g(x), x_3 = x$

Since $0 < |x - a| < \delta = \delta_3$

Then $\left|g(x_3) - \lim_{x\to a} g(x)\right| < \epsilon_3 = \delta_2 \Rightarrow \left|g(x_3) - \lim_{x\to a} g(x)\right| < \epsilon_3 = \delta_2$

Referring back to equation 6, we can then get: $\left|f\big(g(x)\big) - f\left(\lim_{x\to a} g(x)\right)\right| < \epsilon$

5.7 Let's do this again